Dense transcendental power series with easy truncations and a factorial-hard comparison

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Editorial note. Faithful editable TeX transcription of formal author Theorems38/40/44 with complete Section7 proofs, exact sequence/complexity definitions and cited transcendence/functional-equation premises, and full published bibliography. The direct factorial contrast is grouped with the two dense constructions. Original author expressions and conditions preserved without mathematical correction or new proofs; only official CC BY4.0 version used.

Allowed standard prerequisites (editorial).

- Stern generating function identity and transcendence. Exact author Theorem37[12] retained.
- Transcendence of the multiplicative series. Exact author Theorem39[13] retained.
- Zero-radius criterion for formal algebraic series. Exact author Lemma42[28] retained.

Author power-series and complexity conventions

A formal power series is a generalization of a polynomial, where the number of terms can be infinite. Formally, $A = \sum_{i \geq 0} A_i x^i$ with $A_i \in \mathbb{F}$, is a power series in the power series ring $\mathbb{F}[[x]]$. We define the degree d truncation $\text{trunc}(A, d) := \sum_{0 \leq i \leq d} A_i x^i$.

Definition 12 (Algebraic and Transcendental Power Series). A formal power series $f \in \mathbb{C}[[x]]$ is said to be algebraic if there exists a polynomial $g \in \mathbb{C}[x][t]$ such that $g(f) = 0$. Otherwise f is said to be transcendental. For a polynomial $f \in \mathbb{F}[\mathbf{x}]$, $L_{\mathbb{F}}(f)$ is the length of the smallest division-free arithmetic circuit (with only $\{+, -, \times\}$ gates) computing f. Constant-free complexity $\tau(f)$ is the size of the minimal constant-free circuit computing f, with constants in $\{-1, 0, 1\}$. Straight-line programs for integers start at 1 and use $+, -, \times$; $\tau(N)$ is the minimum such length for N. Subscripts are omitted when the field is clear. These are the author conventions from Sections2/5.2.

Definition 36 (The Stern sequence). The sequence $(a_n)_{n\geq 0}$ given by $a_0 = 0, a_1 = 1$, and when $n \geq 1$, by $a_{2n} = a_n$ and $a_{2n+1} = a_n + a_{n+1}$, is called the Stern sequence. Its generating function is $A(x) := \sum a_n x^n$.

Theorem 37 (Lemma2.1 and Theorem2.2 in [12]). If $A(z)$ is the generating function of the Stern sequence, then (1) $A(x^2) = A(x)\frac{x}{x^2+x+1}$; (2) the function $A(x)$ is transcendental.

Multiplicative sequence. The sequence $(f_n)_{n\geq 0}$ is defined as: $f_0 = 1, f_1 = 1, f_2 = -1, f_p = 1$ for all odd primes p and $f_{ab} = f_a f_b$. We look at the corresponding generating function $F(x) := \sum f_n x^n$.

Theorem 39 ([13,Theorem2]). The power series $F(x)$ is transcendental.

Holonomic sequence. A sequence $(h_n)_{n\geq 0}$ is called holonomic if it satisfies a recurrence of the form $a_r(n)h_{n+r} + a_{r-1}(n)h_{n+r-1} + \cdots + a_0(n)h_n = 0$, where a_i are polynomials in n. The corresponding generating function $H(x) := \sum h_n x^n$ is said to be a holonomic function.

Lemma 42 ([28]). If $F(x) = \sum_{n\geq 0} f_n x^n$ is a power series in $\mathbb{C}[[x]]$ and the radius of convergence of $F(x)$ is zero then $F(x)$ is transcendental. This follows directly from Proposition2 in [28], as stated by the authors.

Formal author transcendental-truncation examples

Theorem 38. For the generating function $A(x)$ of the Stern sequence, we have $L(\text{trunc}(A(x), d)) = O(\log^2 d)$.

Theorem 40. For $F(x)$ from the multiplicative sequence in Section7.1.2, we have $L(\text{trunc}(F(x), d)) = O(\log^2 d)$.

Theorem 44. For $F(x) = \sum_{n\geq 0} n! x^n$, if $\tau(\text{trunc}(F(x), d)) = \text{poly}(\log d)$ then $(d!)$ has complexity $\text{poly}(\log d)$.

Complete author Section 7.1.1 proof

Proof of Theorem 38. By using Theorem 37, we obtain that:

$$A(x) = (x^2 + 1)A(x^2) + \frac{A(x^2)}{x}. \tag{3}$$

Suppose $B_d(x) := \text{trunc}(A(x), \lfloor d/2 \rfloor + 1)$. Notice that the degree of $C_d(x) := (x^2+1)B_d(x^2)+B_d(x^2)/x$ is at most $2\lfloor d/2 \rfloor+4$ and $\text{trunc}(C_d(x), d) = \text{trunc}(A(x), d)$. Hence we can compute $\text{trunc}(A(x), d)$ from $C_d(x)$ by subtracting at most 4

monomials, which can be done using $O(\log d)$ gates. Also $B_d(x)$ can be computed from $\text{trunc}(A(x), \lfloor d/2 \rfloor)$ using $O(\log d)$ gates. Hence we obtain the following recurrence:

$$L(\text{trunc}(A(x), d)) \leq L(\text{trunc}(A(x), \lfloor d/2 \rfloor)) + O(\log d).$$

This implies, $L(\text{trunc}(A(x), d)) = O(\log^2 d)$. □

Complete author Section 7.1.2 proof

Proof of Theorem 40. We use the notation $\nu_2(m)$ to denote the highest power of 2 which divides $m \in \mathbb{N}$. We partition the set $[d]$ into $\lfloor \log d \rfloor$ sets $S_0, S_1, S_2, \ldots, S_{\lfloor \log d \rfloor}$ such that $k \in S_i$ iff $\nu_2(k) = i$. We define the set $O_m := \{k : k \leq m \text{ and } k \text{ is odd}\}$. Now, notice that $S_i = \{2^i k : k \in O_{\lfloor d/2^i \rfloor}\}$. For a set $S \in \mathbb{N}$, we define the polynomial $g_S := \sum_{i \in S} x^i$. Observe that:

$$\text{trunc}(F(x), d) = 1 + \sum_{i=1}^{\lfloor \log d \rfloor} (-1)^i g_{S_i}.$$

Trivially, $g_{S_i} = g_{O_{\lfloor d/2^i \rfloor}}(x^{2^i})$. Also notice that $g_{O_m} = g_{[m]} - g_{\lfloor m/2 \rfloor}(x^2)$. Therefore, $L(g_{O_m}) = (\log m)$, which implies that $g_{S_i} = O(\log d)$. Hence, $L(\text{trunc}(F(x), d)) = O(\log^2 d)$. □

Remark 41. Note that, there are power series like $\sum_{i \geq 0} x^{i!}$ which are transcendental and their truncations up to degree d are easy to compute. However, the series is highly sparse and degree-d truncations has only $\text{poly}(\log d)$ monomials, hence the easiness is trivial. The examples we discover in this work are of dense power series.

Complete author Section 7.2 comparison

Consider the holonomic sequence $f_n = (n!)$ defined by $f_0 = 1$ and $f_{n+1} - (n+1)f_n = 0$. Also consider the corresponding generating function $F(x) = \sum_{n \geq 0} n! x^n$. We now show that $F(x)$ is transcendental and that truncation of $F(x)$ is (conditionally) hard to compute.

Corollary 43. The power series $F(x) = \sum_{n \geq 0} n! x^n$ is transcendental.

Proof. It is clear that the radius of convergence of $F(x)$ is zero (follows from the ratio test). Hence Lemma 42 implies that $F(x)$ is transcendental. □

Proof of Theorem 44. We know that $d! x^d = \text{trunc}(F(x), d) - \text{trunc}(F(x), d-1)$. Setting $x = 1$, we conclude. □

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